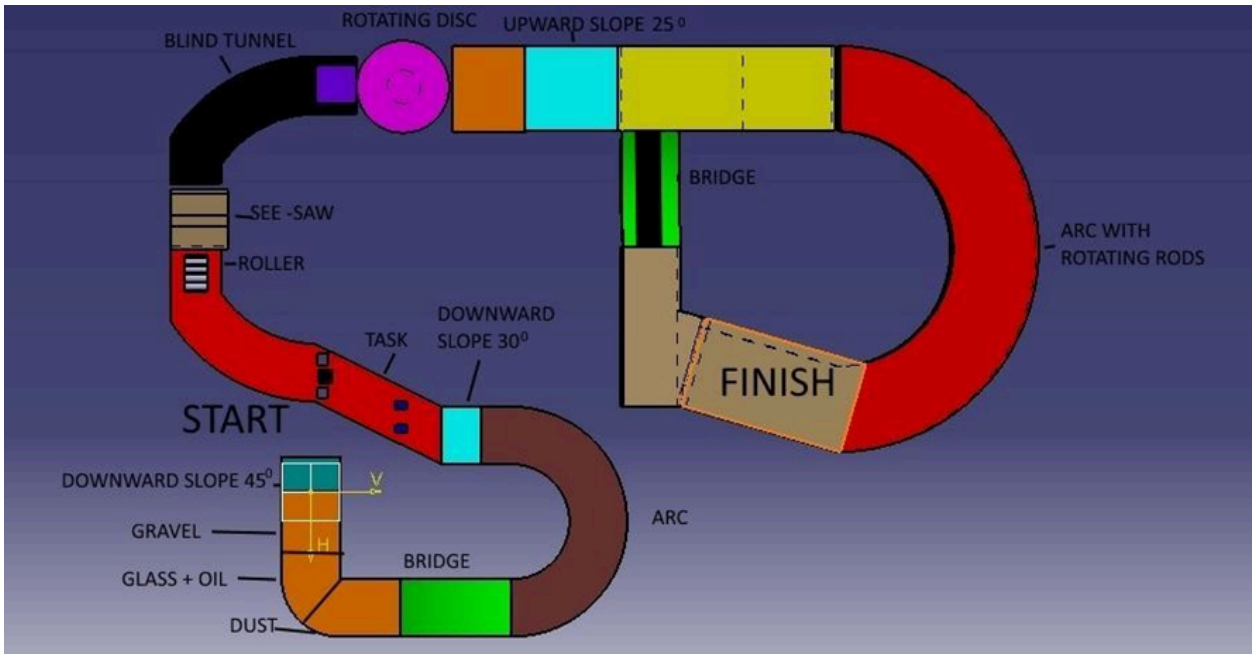
RoboRace



Location: Dr. D. Y. Patil Institute of Technology, Sant Tukaram Nagar, Pimpri Colony, Pune, Pimpri-Chinchwad, Maharashtra 411018

Event Objective:

- This event involves the manufacturing, testing, and racing of a wireless/wired remote-controlled, 2-4-wheeled robot.

Event Structure:

- **The Course:** The track layout will be revealed on the day of the event.
- **The Run:** Each team receives a single attempt to navigate the course.
- **Scoring:** Rankings are based on the shortest time taken to complete the path. The fastest time wins.

Rules of Competition:

1. Eligibility & Team Formation

- **Academic Level:** Participation is strictly limited to **undergraduate students**.
- **Team Composition:** A maximum of **five (5) members** per team is allowed.
- **Inter-Institutional Teams:** Students from different colleges may form a team.

2. Robot Specifications (The Build)

- **Dimensions:** The robot must fit within a box of (L=300mm, B=300mm, H=300mm) at any time during the race. (External controllers are exempt from these dimensions).
- **Power Supply:**
  - The robot must have an onboard supply not exceeding **24V**.
  - **Batteries:** Li-Po, Li-ion, or standard cells are permitted.
  - **Safety:** Batteries must be placed in a **protective shield** in a secure location to prevent damage during collisions or tumbles.
- **Control System:** Wired and Wireless(All Bluetooth modules (3.0, 4.0, 5.0) are applicable).
- **Components:** Motors, tires, and wheels may be directly procured from the market.
- **Integrity:** The vehicle must remain intact. If a robot disintegrates or leaves parts on the track, it will be disqualified.

3. The Track & Arena

- **Terrain:** The course features a combination of wedges, inclinations, and hurdles.
- **Disclosure:** The exact path will be disclosed only at the start of the competition.
- **Disclaimer:** Any provided images are for pictorial representation only; the actual arena layout may differ.

4. Race Procedure & Penalties

- **Trial Format:** Each team is given **one chance** to demonstrate their robot. The best time wins.
- **Checkpoints:** If a robot halts, tumbles, or goes off-course, a member may place it at the **nearest completed checkpoint**. The timer will **not stop** during this process.
- **Time Penalties:**
  - **Robot 30% off-course:** +5 seconds.
  - **Robot 60% off-course:** +10 seconds.
  - **Manual Touch:** +10 seconds per touch.
- **Disqualifications:**
  - Purposefully damaging an opponent's robot or the track.
  - The robot is deemed "unsafe" to run by the coordinator.
  - Touching a robot on invalid grounds.

5. Event Conduct & Logistics

- **Infrastructure:** No power supply will be provided on the track area for charging or operation.
- **Final Authority:** The coordinator's and judges' decisions are **final and binding** to all participants. No protests will be entertained.
- **Right to Amend:** Organizers reserve the right to change rules as they deem fit. Participants are responsible for checking official updates.
- **Race Completion:** Once the race starts, participants must complete the course to receive a recorded time.

Certificate & Registration Policy

- Only teams with fully paid registrations will be eligible for Participation Certificates.
- Certificates will be generated strictly based on the data provided in the Official Google Form. Please ensure names and college details are spelled correctly before submission.
- Due to the high volume of participants, please allow 7–10 working days after the event for the generation and distribution of certificates.
- Digital certificates will be sent to the registered email address of the team lead.
- Important Note: Accuracy in the Google Form is the responsibility of the participant. Re-issuing certificates due to user error may not be possible.

Team and Fee Structure:

- Team size: Maximum 5 Team members (including Team Leader)
- Registration fee: ₹ 350 per team.

Note:

- The arena image provided is for illustrative purposes only. The final track layout, obstacles, and dimensions are subject to change at the organizers' discretion.
- The registration fee is non-refundable.